

PHOENIX Evaluation Study

Phase 2 — Double-Blind Expert Evaluation Survey *Healthcare Professional (POST) Instrument*

Study title	Evaluating the Clinical Quality of an Ontology-Based Multi-Agent Mental Health Support System (PHOENIX)
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Phase	Phase 2 — Blind Evaluation (POST)
Estimated time	Approximately 90–120 minutes (recommended: two sessions of 45–60 min)
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Double-Blind Evaluation Notice

In this phase you will evaluate **ten standardised case vignettes**, each associated with **two anonymised clinical outputs** (labelled **Output A** and **Output B**). The outputs were produced by two different sources — one a human healthcare professional, one an AI reasoning system — but the labels have been **deliberately concealed**. You must not attempt to guess the source.

For each case and each of the five clinical reasoning parts, you will:

1. Read the case vignette and standardised context.
2. Read Output A and Output B independently.
3. Rate *each* output on the relevant clinical quality dimensions (1–9 Likert scale).
4. In Part 5 only: independently classify the HAPA motivational phase *before* reading outputs.

Your judgments are the primary data of this study. Please rate carefully and independently.

This document is the survey blueprint for Qualtrics configuration.
It shows placeholder text for Output A and Output B. In the actual Qualtrics survey, real outputs from Phase 1 are inserted.

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1 Introduction and Participant Information

1.1 Study Overview

This is Phase 2 (POST evaluation) of the PHOENIX master's thesis evaluation study. In Phase 1, five healthcare professionals each independently completed two clinical reasoning cases, producing expert clinical outputs across five tasks (operationalization, initial model, treatment targets, updated model, intervention message). Simultaneously, the PHOENIX AI pipeline processed the same ten cases.

You are one of five independent evaluators in Phase 2. You will assess both sets of outputs (AI and human expert) in a **double-blind format**: you will not know which output was produced by a human and which by the AI system.

1.2 Your Role

- You will evaluate **all ten case vignettes** across **all five clinical reasoning parts**.
- For each case and part, you will see **Output A** and **Output B** — two anonymised outputs.
- You will rate each output independently on **part-specific clinical quality dimensions** using a **1–9 Likert scale**.
- In Part 5, you will also **independently classify the HAPA motivational phase** based on the case information (before reading outputs).

1.3 Ethical Notice and Blinding Protocol

Critical blinding instructions

1. **Do not attempt to identify the source.** The labelling (A/B) is randomised and counterbalanced. No deduction from content is reliable.
2. **Rate each output on its own merits.** Do not compare outputs to each other — rate each independently as if you only had access to that one output.
3. **Use the full 1–9 scale.** Do not cluster your ratings at 7 or above; a score of 9 should reflect genuinely outstanding clinical quality.
4. **Rate all dimensions.** Missing items create missingness in the mixed-effects model; please complete every item.
5. **You may take breaks between cases.** Each case is independent. Complete each part's block of cases in one session if possible.

1.4 Estimated Time and Session Plan

Part	Clinical Task Evaluated	Dimensions	Items	Est. time
1	Operationalization quality	3 per output	60	≈20 min
2	Initial observational model	4 per output	80	≈25 min
3	Treatment-target identification	3 per output	60	≈20 min
4	Updated observational model	2 per output	40	≈15 min
5	Intervention message quality	4 per output + HAPA classification	80+10	≈25 min
Total		16 dimensions	330	≈90–120 min

Recommended sessions:

- **Session 1** (Parts 1–2, all 10 cases): approximately 45 minutes.
- **Session 2** (Parts 3–5, all 10 cases): approximately 50–70 minutes.

2 Rating Scale Guide

All ratings use a 1–9 point Likert scale where:

Score	General Meaning	Example clinical interpretation
1–2	Poor / Absent	Output misses the target entirely or contains major clinical errors
3–4	Below adequate	Partially relevant but with significant omissions or inaccuracies
5	Adequate	Meets minimum clinical standard; workable but unimpressive
6–7	Good	Clinically sound and appropriately detailed
8–9	Excellent	Optimal clinical quality; meets or exceeds expert expectations

Anchors per dimension

Each part provides specific anchors for each dimension. **Rate Output A and Output B completely independently.** Assign ratings based solely on clinical merit, not relative comparison. Both outputs may receive the same score; this is expected and acceptable.

3 Counterbalancing and Blinding Scheme

This section documents the output assignment for researcher reference. **This information is not shown to POST evaluators in the Qualtrics survey.**

Case	Output Source	A	Output Source	B	PRE Respondent	Counterbalance Group
C01	PHOENIX		HCP-PRE-01		HCP-PRE-01	Group I (A=PHOENIX)
C02	PHOENIX		HCP-PRE-01		HCP-PRE-01	Group I (A=PHOENIX)
C03	PHOENIX		HCP-PRE-02		HCP-PRE-02	Group I (A=PHOENIX)
C04	PHOENIX		HCP-PRE-02		HCP-PRE-02	Group I (A=PHOENIX)
C05	PHOENIX		HCP-PRE-03		HCP-PRE-03	Group I (A=PHOENIX)
C06	HCP-PRE-03		PHOENIX		HCP-PRE-03	Group II (B=PHOENIX)
C07	HCP-PRE-04		PHOENIX		HCP-PRE-04	Group II (B=PHOENIX)
C08	HCP-PRE-04		PHOENIX		HCP-PRE-04	Group II (B=PHOENIX)
C09	HCP-PRE-05		PHOENIX		HCP-PRE-05	Group II (B=PHOENIX)
C10	HCP-PRE-05		PHOENIX		HCP-PRE-05	Group II (B=PHOENIX)

The 5/5 split (PHOENIX=A for C01–C05; PHOENIX=B for C06–C10) controls for any systematic bias favoring label A or label B. In analysis, outputs are identified by their true source (PHOENIX vs. HCP) after unblinding.

4 Part 1: Operationalization Quality Evaluation

Part 1 Evaluation Instructions

What was the task? In Phase 1, each respondent was asked to identify and operationalize the key mental health criteria present in the case complaint. They were asked to list 2–6 criteria, each with a short label (2–5 words) and a brief description (1–2 sentences).

Dimensions to rate (1–9 each, rate Output A and B independently):

1. **Criterion accuracy:** Does the output accurately capture the key mental state dimensions described in the complaint? (*1 = major dimensions missed; 9 = all key aspects captured precisely*)
2. **Operationalization quality:** Are the criteria formulated as measurable, time-varying mental health dimensions suitable for repeated self-monitoring? (*1 = not operationalized, merely labels; 9 = all criteria have clear measurement definitions*)
3. **Completeness:** Are all clinically relevant criteria from the complaint represented, without over-inclusion of irrelevant dimensions? (*1 = major omissions or excessive inclusions; 9 = comprehensive and appropriately scoped*)

Qualtrics setup: 10 cases × 2 outputs × 3 dimensions = 60 rating items (Matrix/Likert questions; 1–9 scale). Group cases as blocks.

Time: Approximately 2 minutes per case (≈ 20 minutes total for all 10 cases).

Case C01 — 28-year-old software developer

Sleep onset insomnia, bedtime cognitive hyperarousal, social anhedonia, musculoskeletal somatic tension. Duration: 2 months. Reference case profile for evaluator context only.

[In Qualtrics: insert Phase 1 Output A text for this case/part. Per counterbalancing table: PHOENIX output for C01.]

[In Qualtrics: insert Phase 1 Output B text for this case/part. Per counterbalancing table: HCP-PRE-01 output for C01.]

1a. Criterion accuracy — Output A (circle: 1 ð 2 ð 3 ð 4 ð 5 ð 6 ð 7 ð 8 ð 9)

1 2 3 4 5 6 7 8 9

Poor/Absent

Adequate

Excellent/Optimal

scale.

Rate Output A and Output B independently on this

1b. Criterion accuracy — Output B

(circle: 1 2 3 4 5 6 7 8 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

2a. Operationalization quality — Output A

(circle: 1 2 3 4 5 6 7 8 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

2b. Operationalization quality — Output B

(circle: 1 2 3 4 5 6 7 8 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

3a. Completeness — Output A

(circle: 1 2 3 4 5 6 7 8 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

3b. Completeness — Output B

(circle: 1 2 3 4 5 6 7 8 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

Case C02 — 34-year-old healthcare administrator

Recurrent panic episodes, anticipatory anxiety, situational avoidance, occupational interference. Duration: 3 months.

[PHOENIX output for C02, Part 1.]

[HCP-PRE-01 output for C02, Part 1.]

1a. Criterion accuracy — Output A

(1 2 3 4 5 6 7 8 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

1b. Criterion accuracy — Output B

(1 2 3 4 5 6 7 8 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

2a. Operationalization quality — Output A

(1 2 3 4 5 6 7 8 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

2b. Operationalization quality — Output B

(1 2 3 4 5 6 7 8 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

3a. Completeness — Output A

(1 2 3 4 5 6 7 8 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

3b. Completeness — Output B

(1 2 3 4 5 6 7 8 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

Case C03 — 41-year-old secondary school teacher

Professional burnout, emotional exhaustion, depersonalization, social withdrawal, occupational cognitive difficulties. Duration: 7 months.

[PHOENIX output for C03, Part 1.]

[HCP-PRE-02 output for C03, Part 1.]

1a. Criterion accuracy — Output A (1 ⚪ 2 ⚪ 3 ⚪ 4 ⚪ 5 ⚪ 6 ⚪ 7 ⚪ 8 ⚪ 9)
1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

1b. Criterion accuracy — Output B (1 ⚪ 2 ⚪ 3 ⚪ 4 ⚪ 5 ⚪ 6 ⚪ 7 ⚪ 8 ⚪ 9)
1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

2a. Operationalization quality — Output A (1 ⚪ 2 ⚪ 3 ⚪ 4 ⚪ 5 ⚪ 6 ⚪ 7 ⚪ 8 ⚪ 9)
1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

2b. Operationalization quality — Output B (1 ⚪ 2 ⚪ 3 ⚪ 4 ⚪ 5 ⚪ 6 ⚪ 7 ⚪ 8 ⚪ 9)
1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

3a. Completeness — Output A (1 ⚪ 2 ⚪ 3 ⚪ 4 ⚪ 5 ⚪ 6 ⚪ 7 ⚪ 8 ⚪ 9)
1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

3b. Completeness — Output B (1 ⚪ 2 ⚪ 3 ⚪ 4 ⚪ 5 ⚪ 6 ⚪ 7 ⚪ 8 ⚪ 9)
1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

Case C04 — 63-year-old retired professional

Prolonged grief response, sleep maintenance insomnia, guilt-based rumination, reduced future orientation. Duration: 9 months post-bereavement.

[PHOENIX output for C04, Part 1.]

[HCP-PRE-02 output for C04, Part 1.]

1a. Criterion accuracy — Output A

(1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

1b. Criterion accuracy — Output B

(1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

2a. Operationalization quality — Output A

(1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

2b. Operationalization quality — Output B

(1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

3a. Completeness — Output A

(1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

3b. Completeness — Output B

(1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

Case C05 — 25-year-old postgraduate student

Intrusive harm-related thoughts, compulsive reviewing, experiential avoidance, uncertainty intolerance.
Duration: 20 months.

[PHOENIX output for C05, Part 1.]

[HCP-PRE-03 output for C05, Part 1.]

1a. Criterion accuracy — Output A (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)
1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

1b. Criterion accuracy — Output B (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)
1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

2a. Operationalization quality — Output A (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)
1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

2b. Operationalization quality — Output B (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)
1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

3a. Completeness — Output A (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)
1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

3b. Completeness — Output B (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)
1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

Case C06 — 32-year-old marketing professional

Professional performance anxiety, post-event processing, advancement avoidance, chronic occupational anxiety. Duration: >3 years.

[HCP-PRE-03 output for C06, Part 1. (Note to researcher: counterbalanced — A=HCP for C06–C10.)]

[PHOENIX output for C06, Part 1.]

1a. Criterion accuracy — Output A (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)
 1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
Rate Output A and Output B independently on this scale.

1b. Criterion accuracy — Output B (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)
 1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
Rate Output A and Output B independently on this scale.

2a. Operationalization quality — Output A (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)
 1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
Rate Output A and Output B independently on this scale.

2b. Operationalization quality — Output B (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)
 1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
Rate Output A and Output B independently on this scale.

3a. Completeness — Output A (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)
 1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
Rate Output A and Output B independently on this scale.

3b. Completeness — Output B (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)
 1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
Rate Output A and Output B independently on this scale.

Case C07 — 46-year-old accounts manager

Persistent low mood, anergia, social isolation, hopelessness, anhedonia. Duration: 14 months.

[HCP-PRE-04 output for C07, Part 1.]

[PHOENIX output for C07, Part 1.]

1a. Criterion accuracy — Output A

(1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

1b. Criterion accuracy — Output B

(1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

2a. Operationalization quality — Output A

(1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

2b. Operationalization quality — Output B

(1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

3a. Completeness — Output A

(1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

3b. Completeness — Output B

(1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

Case C08 — 38-year-old project manager

ADHD-related executive dysfunction, emotional dysregulation, interpersonal conflict, shame response. Childhood onset.

[HCP-PRE-04 output for C08, Part 1.]

[PHOENIX output for C08, Part 1.]

1a. Criterion accuracy — Output A

(1 ⚐ 2 ⚐ 3 ⚐ 4 ⚐ 5 ⚐ 6 ⚐ 7 ⚐ 8 ⚐ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

1b. Criterion accuracy — Output B

(1 ⚐ 2 ⚐ 3 ⚐ 4 ⚐ 5 ⚐ 6 ⚐ 7 ⚐ 8 ⚐ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

2a. Operationalization quality — Output A

(1 ⚐ 2 ⚐ 3 ⚐ 4 ⚐ 5 ⚐ 6 ⚐ 7 ⚐ 8 ⚐ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

2b. Operationalization quality — Output B

(1 ⚐ 2 ⚐ 3 ⚐ 4 ⚐ 5 ⚐ 6 ⚐ 7 ⚐ 8 ⚐ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

3a. Completeness — Output A

(1 ⚐ 2 ⚐ 3 ⚐ 4 ⚐ 5 ⚐ 6 ⚐ 7 ⚐ 8 ⚐ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

3b. Completeness — Output B

(1 ⚐ 2 ⚐ 3 ⚐ 4 ⚐ 5 ⚐ 6 ⚐ 7 ⚐ 8 ⚐ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

Case C09 — 27-year-old sales representative

Emotional dysregulation, impulsivity, interpersonal instability, identity disturbance. Duration: 3 years.

[HCP-PRE-05 output for C09, Part 1.]

[PHOENIX output for C09, Part 1.]

1a. Criterion accuracy — Output A (1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this scale.

1b. Criterion accuracy — Output B (1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this scale.

2a. Operationalization quality — Output A (1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this scale.

2b. Operationalization quality — Output B (1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this scale.

3a. Completeness — Output A (1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this scale.

3b. Completeness — Output B (1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this scale.

Case C10 — 52-year-old operations director

Chronic occupational stress, sleep maintenance insomnia, somatic stress symptoms, evening alcohol coping. Duration: 9 months.

[HCP-PRE-05 output for C10, Part 1.]

[PHOENIX output for C10, Part 1.]

1a. Criterion accuracy — Output A

(1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

1b. Criterion accuracy — Output B

(1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

2a. Operationalization quality — Output A

(1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

2b. Operationalization quality — Output B

(1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

3a. Completeness — Output A

(1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

3b. Completeness — Output B

(1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

5 Part 2: Initial Observational Model Evaluation

Part 2 Evaluation Instructions

What was the task? Respondents were asked to propose 3–6 modifiable predictor variables that could be measured daily alongside the criteria identified in Part 1, and to describe how each predictor connects to the criteria network.

Dimensions to rate (1–9 each, rate A and B independently):

1. **Clinical appropriateness:** Are the proposed predictors clinically plausible, evidence-supported risk/maintenance factors for this presentation? (*1 = implausible; 9 = all strongly evidence-based*)
2. **Network validity:** Do the proposed predictor-to-criterion connections make clinical sense? (*1 = connections clinically unjustifiable; 9 = all connections clinically sound and specific*)
3. **EMA feasibility:** Can the proposed predictors realistically be monitored via daily ecological momentary assessment? (*1 = not self-monitorable; 9 = all fully feasible for once/twice-daily EMA*)
4. **Intervention potential:** Do the proposed predictors represent genuinely modifiable targets that could be the focus of a digital intervention? (*1 = no modifiable levers; 9 = multiple high-leverage, modifiable predictors*)

Context shown to evaluators: In addition to the vignette, evaluators see the standardised criteria list from Part 1 (as context for the model they are evaluating).

Time: Approximately 2.5 minutes per case ($\approx$ 25 minutes total).

Case C01 — 28-year-old software developer

Standardised criteria (context): CR-1: Sleep onset insomnia CR-2: Cognitive hyperarousal CR-3: Social anhedonia CR-4: Musculoskeletal tension

[PHOENIX output for C01, Part 2.]

[HCP-PRE-01 output for C01, Part 2.]

1a. Clinical appropriateness — Output A (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

1b. Clinical appropriateness — Output B (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

2a. Network validity — Output A (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

2b. Network validity — Output B (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

3a. EMA feasibility — Output A (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

3b. EMA feasibility — Output B (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

4a. Intervention potential — Output A (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

4b. Intervention potential — Output B (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

[Cases C02–C10 follow the same format with their respective criteria context and outputs.]

Case C02 — 34-year-old healthcare administrator

Standardised criteria: CR-1: Recurrent panic episodes CR-2: Anticipatory anxiety CR-3: Situational avoidance CR-4: Occupational interference

[PHOENIX output for C02, Part 2.]

[HCP-PRE-01 output for C02, Part 2.]

1a. Clinical appropriateness — Output A (1–9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

1b. Clinical appropriateness — Output B (1–9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

2a. Network validity — Output A (1–9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

2b. Network validity — Output B (1–9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

3a. EMA feasibility — Output A (1–9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

3b. EMA feasibility — Output B (1–9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

4a. Intervention potential — Output A

(1–9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

4b. Intervention potential — Output B

(1–9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

Case C03 — 41-year-old secondary school teacher

Standardised criteria: CR-1: Emotional exhaustion CR-2: Professional depersonalization CR-3: Social withdrawal CR-4: Occupational cognitive impairment

[PHOENIX output for C03, Part 2.]

[HCP-PRE-02 output for C03, Part 2.]

1a–4b. Rate clinical appropriateness, network validity, EMA feasibility, intervention potential for Outputs A and B (8 items, 1–9 each).

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

Case C04 — 63-year-old retired professional

Standardised criteria: CR-1: Prolonged grief response CR-2: Sleep maintenance insomnia CR-3: Guilt-based rumination CR-4: Reduced future orientation

[PHOENIX output for C04, Part 2.]

[HCP-PRE-02 output for C04, Part 2.]

1a–4b. Rate clinical appropriateness, network validity, EMA feasibility, intervention potential for Outputs A and B (8 items, 1–9 each).

1 2 3 4 5 6 7 8 9
 Poor/Absent Adequate Excellent/Optimal
 scale.

Rate Output A and Output B independently on this scale.

Case C05 — 25-year-old postgraduate student

Standardised criteria: CR-1: Intrusive harm-related thoughts CR-2: Compulsive reviewing behaviour CR-3: Experiential avoidance CR-4: Uncertainty intolerance

[PHOENIX output for C05, Part 2.]

[HCP-PRE-03 output for C05, Part 2.]

1a–4b. Rate all four dimensions for Outputs A and B (8 items, 1–9 each).

1 2 3 4 5 6 7 8 9
 Poor/Absent Adequate Excellent/Optimal
 scale.

Rate Output A and Output B independently on this scale.

Case C06 — 32-year-old marketing professional

Standardised criteria: CR-1: Social performance anxiety CR-2: Post-event processing CR-3: Advancement avoidance CR-4: Anticipatory cognitive elaboration

[HCP-PRE-03 output for C06, Part 2. (A=HCP for C06–C10.)]

[PHOENIX output for C06, Part 2.]

1a–4b. Rate all four dimensions for Outputs A and B (8 items, 1–9 each).

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

Case C07 — 46-year-old accounts manager

Standardised criteria: CR-1: Persistent low mood CR-2: Anergia and psychomotor slowing CR-3: Social isolation CR-4: Hopelessness and negative schema

[HCP-PRE-04 output for C07, Part 2.]

[PHOENIX output for C07, Part 2.]

1a–4b. Rate all four dimensions for Outputs A and B (8 items, 1–9 each).

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

Case C08 — 38-year-old project manager

Standardised criteria: CR-1: Executive dysfunction CR-2: Emotional dysregulation (ADHD-related) CR-3: Interpersonal conflict CR-4: Task-completion shame cycle

[HCP-PRE-04 output for C08, Part 2.]

[PHOENIX output for C08, Part 2.]

1a–4b. Rate all four dimensions for Outputs A and B (8 items, 1–9 each).

1	2	3	4	5	6	7	8	9	Rate Output A and Output B independently on this scale.
Poor/Absent			Adequate		Excellent/Optimal				

Case C09 — 27-year-old sales representative

Standardised criteria: CR-1: Affective instability CR-2: Impulsive behaviour CR-3: Unstable interpersonal schema CR-4: Identity diffusion

[HCP-PRE-05 output for C09, Part 2.]

[PHOENIX output for C09, Part 2.]

1a–4b. Rate all four dimensions for Outputs A and B (8 items, 1–9 each).

1	2	3	4	5	6	7	8	9	Rate Output A and Output B independently on this scale.
Poor/Absent			Adequate		Excellent/Optimal				

Case C10 — 52-year-old operations director

Standardised criteria: CR-1: Chronic occupational stress response CR-2: Sleep maintenance insomnia CR-3: Somatic stress expression CR-4: Alcohol use as coping

[HCP-PRE-05 output for C10, Part 2.]

[PHOENIX output for C10, Part 2.]

1a–4b. Rate all four dimensions for Outputs A and B (8 items, 1–9 each).

1	2	3	4	5	6	7	8	9	<i>Rate Output A and Output B independently on this scale.</i>
<i>Poor/Absent</i>			<i>Adequate</i>			<i>Excellent/Optimal</i>			

6 Part 3: Treatment-Target Identification Evaluation

Part 3 Evaluation Instructions

What was the task? Respondents were shown the bipartite network (criteria and predictors with edge strengths derived from 21-day monitoring data) and asked to rank the top 2–4 predictors as treatment targets, in order of priority.

Dimensions to rate (1–9 each, rate A and B independently):

1. **Clinical priority:** Are the identified treatment targets the most clinically important ones given this presentation and the monitoring data? (*1 = clearly suboptimal targets; 9 = optimal targets identified*)
2. **Evidence alignment:** Does the selection and ranking reflect evidence-based treatment priorities for this type of presentation? (*1 = not evidence-grounded; 9 = strongly aligned with evidence-based guidelines*)
3. **Rank coherence:** Is the ordering of targets (1st, 2nd, etc.) clinically defensible, with higher-ranked items having greater expected impact? (*1 = arbitrary or reversed ordering; 9 = perfectly coherent clinical prioritization*)

Context provided: Evaluators see the bipartite network diagram and 21-day monitoring summary for each case.

Time: Approximately 2 minutes per case (≈ 20 minutes total).

Case C01 — 28-year-old software developer

Network context: Predictors: P1 Evening screen time (Strong $\rightarrow$ CR-1, CR-2); P2 Daily task incompleteness (Moderate $\rightarrow$ CR-2); P3 Pre-sleep offloading (Strong $\rightarrow$ CR-2, Moderate $\rightarrow$ CR-1); P4 Social engagement (Strong $\rightarrow$ CR-3); P5 Physical activity (Moderate $\rightarrow$ CR-1, CR-4).

[PHOENIX ranked target list for C01, Part 3.]

[HCP-PRE-01 ranked target list for C01, Part 3.]

1a. Clinical priority — Output A

(1 ù 2 ù 3 ù 4 ù 5 ù 6 ù 7 ù 8 ù 9)

1 2 3 4 5 6 7 8 9

Poor/Absent

Adequate

Excellent/Optimal

scale.

Rate Output A and Output B independently on this

1b. Clinical priority — Output B

(1 ⚑ 2 ⚑ 3 ⚑ 4 ⚑ 5 ⚑ 6 ⚑ 7 ⚑ 8 ⚑ 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

2a. Evidence alignment — Output A

(1 ⚑ 2 ⚑ 3 ⚑ 4 ⚑ 5 ⚑ 6 ⚑ 7 ⚑ 8 ⚑ 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

2b. Evidence alignment — Output B

(1 ⚑ 2 ⚑ 3 ⚑ 4 ⚑ 5 ⚑ 6 ⚑ 7 ⚑ 8 ⚑ 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

3a. Rank coherence — Output A

(1 ⚑ 2 ⚑ 3 ⚑ 4 ⚑ 5 ⚑ 6 ⚑ 7 ⚑ 8 ⚑ 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

3b. Rank coherence — Output B

(1 ⚑ 2 ⚑ 3 ⚑ 4 ⚑ 5 ⚑ 6 ⚑ 7 ⚑ 8 ⚑ 9)

1 2 3 4 5 6 7 8 9

Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

Case C02 — 34-year-old healthcare administrator

Network context: P1 Avoidance events (Strong → CR-3, CR-2); P2 Safety behaviour frequency (Strong → CR-2, Moderate → CR-1); P3 Interoceptive focus duration (Strong → CR-2, CR-1); P4 Planned exposure attempts (Strong → CR-3); P5 Daily structure adherence (Moderate → CR-4).

[PHOENIX output for C02, Part 3.]

[HCP-PRE-01 output for C02, Part 3.]

1a–3b. Rate clinical priority, evidence alignment, rank coherence for A and B (6 items, 1–9).

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

Case C03 — 41-year-old secondary school teacher

Network context: P1 Work-life boundary adherence (Strong → CR-1, CR-2); P2 Recovery activity engagement (Strong → CR-1, Moderate → CR-3); P3 Social contact initiation (Strong → CR-3); P4 Professional mastery moments (Moderate → CR-2, CR-4); P5 Sleep quality (Moderate → CR-1, CR-4).

[PHOENIX output for C03, Part 3.]

[HCP-PRE-02 output for C03, Part 3.]

1a–3b. Rate clinical priority, evidence alignment, rank coherence for A and B (6 items, 1–9).

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal

Rate Output A and Output B independently on this scale.

Cases C04–C10

Each remaining case (C04–C10) follows the identical format: vignette with network context summary, Output A placeholder, Output B placeholder, and 6 rating items (clinical priority / evidence alignment / rank coherence for A and B). See Appendix A for the full network context for each case.

7 Part 4: Updated Observational Model Evaluation

Part 4 Evaluation Instructions

What was the task? Given the identified treatment targets from Part 3, respondents were asked to select 2–4 predictors to retain in the next monitoring cycle and optionally provide a 1-sentence operationalization note.

Dimensions to rate (1–9 each, rate A and B independently):

1. **Target alignment:** Does the updated measurement plan directly reflect the identified treatment targets? Do the retained predictors logically follow from the treatment target selection in Part 3? (*1 = misaligned; 9 = perfectly coherent with treatment targets*)
2. **Measurement selection quality:** Are the retained predictors optimal choices for tracking progress on the treatment targets? Are operationalization notes (if provided) clinically appropriate? (*1 = poor measurement choices; 9 = optimal and actionable measurement plan*)

Context: Evaluators see the standardised treatment targets from Step 03 reference run alongside the outputs.

Time: Approximately 1.5 minutes per case (≈ 15 minutes total).

Case C01 — 28-year-old software developer

Reference treatment targets: (1) Evening screen time duration; (2) Pre-sleep cognitive offloading; (3) Aerobic physical activity.

[PHOENIX updated model selection for C01, Part 4.]

[HCP-PRE-01 updated model selection for C01, Part 4.]

1a. Target alignment — Output A

(1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

1b. Target alignment — Output B

(1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

2a. Measurement selection quality — Output A (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

2b. Measurement selection quality — Output B (1 ⚡ 2 ⚡ 3 ⚡ 4 ⚡ 5 ⚡ 6 ⚡ 7 ⚡ 8 ⚡ 9)

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

Case C02 — 34-year-old healthcare administrator

Reference treatment targets: (1) Planned exposure attempts; (2) Safety behaviour frequency; (3) Interoceptive focus duration.

[PHOENIX output for C02, Part 4.]

[HCP-PRE-01 output for C02, Part 4.]

1a–2b. Rate target alignment and measurement selection quality for A and B (4 items, 1–9).

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

Case C03 — 41-year-old secondary school teacher

Reference treatment targets: (1) Work-life boundary adherence; (2) Recovery activity engagement; (3) Professional mastery moments.

[PHOENIX output for C03, Part 4.]

[HCP-PRE-02 output for C03, Part 4.]

1a–2b. Rate target alignment and measurement selection quality for A and B (4 items, 1–9).

1	2	3	4	5	6	7	8	9	<i>Rate Output A and Output B independently on this scale.</i>
<i>Poor/Absent</i>			<i>Adequate</i>		<i>Excellent/Optimal</i>				

Cases C04–C10

Each case follows the same 4-item format. For cases C04–C10, reference treatment targets are listed in Appendix A.

8 Part 5: Tailored Intervention Message Evaluation

Part 5 Evaluation Instructions

What was the task? Respondents were asked to: (a) classify the person's HAPA motivational phase (pre-intentional / intentional / action-maintenance), and (b) write a brief 2–3 sentence personalized digital coaching message appropriate to that phase.

Step A — Independent HAPA phase classification (before reading outputs):

- For each case, **classify the HAPA motivational phase** based solely on the case vignette and monitoring context.
- Your classification will be used to calculate inter-rater agreement (Cohen's κ) across all POST evaluators and the Phase 1 expert respondents.
- **Do not read Outputs A or B before completing this classification.**

Step B — Rating Outputs A and B (1–9 each, rate independently):

1. **HAPA phase appropriateness:** Does the coaching message address the correct motivational phase? Is it pitched at the right level for where the person is in their change process? (*1 = wrong phase; 9 = perfectly calibrated to motivational stage*)
2. **Message tailoring:** Is the message genuinely personalized to this specific case (references their specific situation, complaint, monitoring patterns)? (*1 = generic; 9 = highly personalized and case-specific*)
3. **Actionability:** Does the message provide concrete, actionable guidance the person can act on within 24–48 hours? (*1 = vague platitudes; 9 = specific, immediately actionable steps*)
4. **Professional tone:** Is the language appropriately supportive, non-judgmental, and professional for a digital health context? (*1 = inappropriate; 9 = exemplary professional and supportive tone*)

Time: Approximately 2.5 minutes per case (≈ 25 minutes total).

Case C01 — 28-year-old software developer

Monitoring context for HAPA classification: 21-day data shows: Screen time reduction attempts: 2/21 evenings. No structured sleep hygiene practice initiated. However: task completion tracking started Week 3. Participant reported “trying a bit more” in final week. Phase indicators: partial intention formation, no systematic action plan yet.

HAPA Phase Classification (Multiple Choice)

Based on the case information only (before reading outputs), select the most appropriate motivational phase:

- ☐ **Pre-Intentional** — person has not yet formed a clear behavioural intention
- ☐ **Intentional** — person intends to change but has not yet started acting
- ☐ **Action-Maintenance** — person is actively engaged in behaviour change

This is your independent clinical judgment — do not be influenced by Outputs A or B.

[PHOENIX HAPA classification + coaching message for C01.]

[HCP-PRE-01 HAPA classification + coaching message for C01.]

1a. HAPA phase appropriateness — Output A (1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)
 1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

1b. HAPA phase appropriateness — Output B (1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)
 1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

2a. Message tailoring — Output A (1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)
 1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

2b. Message tailoring — Output B (1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)
 1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

3a. Actionability — Output A (1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)
 1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

3b. Actionability — Output B (1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)
 1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

4a. Professional tone — Output A (1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)
 1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

4b. Professional tone — Output B (1 ⚭ 2 ⚭ 3 ⚭ 4 ⚭ 5 ⚭ 6 ⚭ 7 ⚭ 8 ⚭ 9)
 1 2 3 4 5 6 7 8 9 *Rate Output A and Output B independently on this*
Poor/Absent Adequate Excellent/Optimal
scale.

Case C02 — 34-year-old healthcare administrator

Monitoring context: Participant has identified avoiding situations as a problem (intention awareness). Has not yet started planned exposure. Attendance drop to 60% suggests avoidance escalating. Phase indicators: intentional (knows what needs to change, has not yet acted systematically).

HAPA Phase Classification (Multiple Choice)

Based on the case information only (before reading outputs), select the most appropriate motivational phase:

- ☐ **Pre-Intentional** — person has not yet formed a clear behavioural intention
- ☐ **Intentional** — person intends to change but has not yet started acting
- ☐ **Action-Maintenance** — person is actively engaged in behaviour change

This is your independent clinical judgment — do not be influenced by Outputs A or B.

[PHOENIX output for C02, Part 5.]

[HCP-PRE-01 output for C02, Part 5.]

1a–4b. Rate HAPA phase appropriateness, message tailoring, actionability, professional tone for A and B (8 items, 1–9).

1 2 3 4 5 6 7 8 9
 Poor/Absent Adequate Excellent/Optimal
 scale.

Rate Output A and Output B independently on this scale.

Case C03 — 41-year-old secondary school teacher

Monitoring context: Recovery activities almost absent (1.2/week). No deliberate boundary-setting attempts in monitoring data. Social withdrawal has deepened. Phase indicators: pre-intentional (person not yet motivated to change specific behaviours; complaint = burnout overwhelm).

HAPA Phase Classification (Multiple Choice)

Based on the case information only (before reading outputs), select the most appropriate motivational phase:

- ☐ **Pre-Intentional** — person has not yet formed a clear behavioural intention
- ☐ **Intentional** — person intends to change but has not yet started acting
- ☐ **Action-Maintenance** — person is actively engaged in behaviour change

This is your independent clinical judgment — do not be influenced by Outputs A or B.

[PHOENIX output for C03, Part 5.]

[HCP-PRE-02 output for C03, Part 5.]

1a–4b. Rate all four dimensions for A and B (8 items, 1–9).

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

Case C04 — 63-year-old retired professional

Monitoring context: Meaning-making activity tried 4/21 days (action-maintenance pattern emerging). Sleep routine shows Week 3 improvement. However future-planning activities remain absent. Phase indicators: mixed intentional/early action-maintenance.

HAPA Phase Classification (Multiple Choice)

Based on the case information only (before reading outputs), select the most appropriate motivational phase:

- ☐ **Pre-Intentional** — person has not yet formed a clear behavioural intention
- ☐ **Intentional** — person intends to change but has not yet started acting
- ☐ **Action-Maintenance** — person is actively engaged in behaviour change

This is your independent clinical judgment — do not be influenced by Outputs A or B.

[PHOENIX output for C04, Part 5.]

[HCP-PRE-02 output for C04, Part 5.]

1a–4b. Rate all four dimensions for A and B (8 items, 1–9).

1 2 3 4 5 6 7 8 9
Poor/Absent Adequate Excellent/Optimal
scale.

Rate Output A and Output B independently on this

Case C05 — 25-year-old postgraduate student

Monitoring context: No ERP practice initiated. Compulsions still at full frequency. No deliberate delay attempts logged. Phase indicators: pre-intentional (person still in distress cycle with no action intention formulated for avoidance reduction).

HAPA Phase Classification (Multiple Choice)

Based on the case information only (before reading outputs), select the most appropriate motivational phase:

- ☐ **Pre-Intentional** — person has not yet formed a clear behavioural intention
- ☐ **Intentional** — person intends to change but has not yet started acting
- ☐ **Action-Maintenance** — person is actively engaged in behaviour change

This is your independent clinical judgment — do not be influenced by Outputs A or B.

[PHOENIX output for C05, Part 5.]

[HCP-PRE-03 output for C05, Part 5.]

1a–4b. Rate all four dimensions for A and B (8 items, 1–9).

1	2	3	4	5	6	7	8	9
<i>Poor/Absent</i>		<i>Adequate</i>		<i>Excellent/Optimal</i>				

Rate Output A and Output B independently on this scale.

Case C06 — 32-year-old marketing professional

Monitoring context: Post-event processing still high (avg 87 min). Participant has sought information about social anxiety (intentional awareness). One opportunity deliberately accepted Week 3. Phase indicators: transitional intentional/early action-maintenance.

HAPA Phase Classification (Multiple Choice)

Based on the case information only (before reading outputs), select the most appropriate motivational phase:

- ☐ **Pre-Intentional** — person has not yet formed a clear behavioural intention
- ☐ **Intentional** — person intends to change but has not yet started acting
- ☐ **Action-Maintenance** — person is actively engaged in behaviour change

This is your independent clinical judgment — do not be influenced by Outputs A or B.

[HCP-PRE-03 output for C06, Part 5. (A=HCP for C06–C10.)]

[PHOENIX output for C06, Part 5.]

1a–4b. Rate all four dimensions for A and B (8 items, 1–9).

1	2	3	4	5	6	7	8	9	
Poor/Absent			Adequate			Excellent/Optimal			Rate Output A and Output B independently on this scale.

Case C07 — 46-year-old accounts manager

Monitoring context: Activity scheduling completion at 28%. Has not initiated social contact. No structured routine. Phase indicators: intentional (aware of need for activation but not acting consistently).

HAPA Phase Classification (Multiple Choice)

Based on the case information only (before reading outputs), select the most appropriate motivational phase:

- ☐ **Pre-Intentional** — person has not yet formed a clear behavioural intention
- ☐ **Intentional** — person intends to change but has not yet started acting
- ☐ **Action-Maintenance** — person is actively engaged in behaviour change

This is your independent clinical judgment — do not be influenced by Outputs A or B.

[HCP-PRE-04 output for C07, Part 5.]

[PHOENIX output for C07, Part 5.]

1a–4b. Rate all four dimensions for A and B (8 items, 1–9).

1	2	3	4	5	6	7	8	9	
Poor/Absent			Adequate			Excellent/Optimal			Rate Output A and Output B independently on this scale.

Case C08 — 38-year-old project manager

Monitoring context: Task structuring tool used inconsistently (1.8/day). Has begun using timer app in Week 3 (action-maintenance). Relationship conflict still high. Phase indicators: action-maintenance for executive function; pre-intentional for relationship domain.

HAPA Phase Classification (Multiple Choice)

Based on the case information only (before reading outputs), select the most appropriate motivational phase:

- ☐ **Pre-Intentional** — person has not yet formed a clear behavioural intention
- ☐ **Intentional** — person intends to change but has not yet started acting
- ☐ **Action-Maintenance** — person is actively engaged in behaviour change

This is your independent clinical judgment — do not be influenced by Outputs A or B.

[HCP-PRE-04 output for C08, Part 5.]

[PHOENIX output for C08, Part 5.]

1a–4b. Rate all four dimensions for A and B (8 items, 1–9).

1	2	3	4	5	6	7	8	9
Poor/Absent			Adequate			Excellent/Optimal		

Rate Output A and Output B independently on this scale.

Case C09 — 27-year-old sales representative

Monitoring context: 2.1 impulsive actions/day, no delay-and-reappraise strategy used. High rejection sensitivity. No DBT-type skills practice logged. Phase indicators: pre-intentional (not yet recognising need for systematic self-regulation skills development).

HAPA Phase Classification (Multiple Choice)

Based on the case information only (before reading outputs), select the most appropriate motivational phase:

- ☐ **Pre-Intentional** — person has not yet formed a clear behavioural intention
- ☐ **Intentional** — person intends to change but has not yet started acting
- ☐ **Action-Maintenance** — person is actively engaged in behaviour change

This is your independent clinical judgment — do not be influenced by Outputs A or B.

[HCP-PRE-05 output for C09, Part 5.]

[PHOENIX output for C09, Part 5.]

1a–4b. Rate all four dimensions for A and B (8 items, 1–9).

1	2	3	4	5	6	7	8	9	
Poor/Absent			Adequate			Excellent/Optimal			Rate Output A and Output B independently on this scale.

Case C10 — 52-year-old operations director

Monitoring context: Alcohol reduction attempted twice. Pre-sleep disengagement tried 3/21 evenings. Work demands still high but participant reports “knowing I need to change.” Phase indicators: intentional (aware, partially motivated, not yet sustaining change).

HAPA Phase Classification (Multiple Choice)

Based on the case information only (before reading outputs), select the most appropriate motivational phase:

- ☐ **Pre-Intentional** — person has not yet formed a clear behavioural intention
- ☐ **Intentional** — person intends to change but has not yet started acting
- ☐ **Action-Maintenance** — person is actively engaged in behaviour change

This is your independent clinical judgment — do not be influenced by Outputs A or B.

[HCP-PRE-05 output for C10, Part 5.]

[PHOENIX output for C10, Part 5.]

1a–4b. Rate all four dimensions for A and B (8 items, 1–9).

1	2	3	4	5	6	7	8	9	
Poor/Absent			Adequate			Excellent/Optimal			Rate Output A and Output B independently on this scale.

A Appendix A: Reference Data for Cases C04–C10 (Parts 3 and 4)

Part 3 Network Context: Cases C04–C10

Case	Network predictor summary (for evaluator context)
C04	P1 Meaning-making activity (Strong → CR-1, CR-4); P2 Sleep routine consistency (Strong → CR-2); P3 Social support engagement (Moderate → CR-1, CR-3); P4 Self-compassion practices (Strong → CR-3); P5 Future-self activity planning (Strong → CR-4)
C05	P1 Compulsion delay practice (Strong → CR-2, CR-1); P2 Avoidance scope management (Strong → CR-3, CR-1); P3 Uncertainty tolerance behaviour (Strong → CR-4, CR-3); P4 Intrusion logging and defusion (Moderate → CR-1); P5 Planned task exposure (Strong → CR-2)
C06	P1 Graduated professional exposure (Strong → CR-3, CR-1); P2 Post-event processing duration (Strong → CR-2); P3 Anticipatory anxiety coping self-talk (Moderate → CR-1, CR-4); P4 Avoidance decision logging (Strong → CR-3); P5 Physical preparation routines (Weak → CR-1)
C07	P1 Scheduled activity completion (Strong → CR-1, CR-2); P2 Sleep schedule regularity (Strong → CR-2, Moderate → CR-1); P3 Social contact initiation (Strong → CR-3); P4 Mastery and pleasure tracking (Moderate → CR-1, CR-4); P5 Rumination interruption practice (Moderate → CR-4)
C08	P1 External task structuring tool use (Strong → CR-1); P2 Intentional listening behaviour (Strong → CR-3); P3 Task completion reinforcement (Moderate → CR-4); P4 Pre-task planning minutes (Moderate → CR-1); P5 Emotional regulation before response (Moderate → CR-2, CR-3)
C09	P1 Emotion regulation skill application (Strong → CR-1, CR-2); P2 Impulse delay behaviour (Strong → CR-2, Moderate → CR-3); P3 Rejection appraisal reframing (Strong → CR-3); P4 Identity anchoring practice (Moderate → CR-4); P5 Interpersonal conflict reflection (Moderate → CR-3)
C10	P1 Pre-sleep work disengagement (Strong → CR-2, CR-1); P2 Evening alcohol units (Strong → CR-4, CR-2); P3 Daytime stress regulation (Strong → CR-1, Moderate → CR-3); P4 Physical activity frequency (Moderate → CR-1, CR-3); P5 Nocturnal cognitive arousal logging (Strong → CR-2)

Part 4 Reference Treatment Targets: Cases C04–C10

Standardised top treatment targets (from Step 03 reference run)	
Case	
C04	(1) Structured meaning-making activity; (2) Sleep routine consistency; (3) Self-compassion practices
C05	(1) Compulsion delay and resistance practice; (2) Avoidance scope management; (3) Uncertainty tolerance behaviour
C06	(1) Graduated professional exposure frequency; (2) Post-event processing duration; (3) Avoidance decision logging
C07	(1) Daily activity scheduling and completion; (2) Sleep schedule regularity; (3) Social contact initiation
C08	(1) External task structuring tool use; (2) Intentional listening behaviour; (3) Task completion reinforcement
C09	(1) Emotion regulation skill application; (2) Impulse delay behaviour; (3) Rejection appraisal reframing
C10	(1) Pre-sleep work disengagement routine; (2) Evening alcohol units; (3) Daytime stress regulation

Qualtrics Data Export Specification

The following variable naming convention should be used in Qualtrics for clean data export and automated analysis:

Variable name	Description
participant_ID	POST evaluator code (HCP-POST-01 through HCP-POST-05)
case_ID	Case identifier (C01–C10)
part	Study part (1–5)
output_label	A or B (blinded label)
dimension	Rating dimension name (e.g., criterion_accuracy)
rating	Integer rating 1–9
hapa_class	POST rater HAPA classification (Part 5 only; 1=pre-intentional, 2=intentional, 3=action-maintenance)